

# **Semantics of Comparative Constructions**

## Class 1: Gradable adjectives & degrees

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## Part 1

# Introductory remarks

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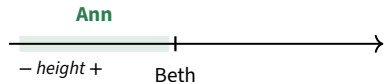
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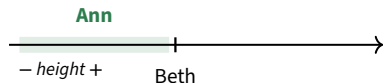
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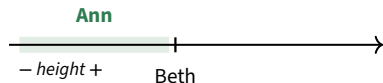
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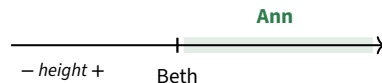
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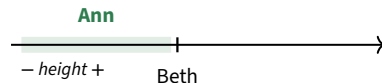
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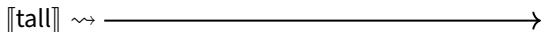
▷ **Our goal in this course:** develop a compositional semantic analysis of these constructions

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- This course is an introduction to analyses of comparatives based on the notion of **degrees**.

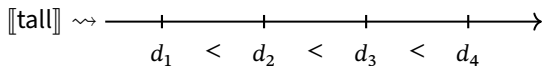
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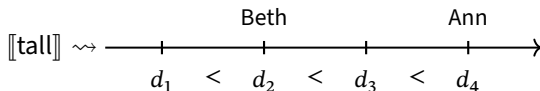
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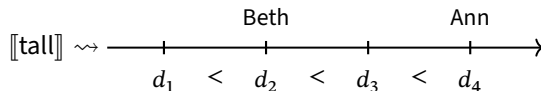
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- Degrees go beyond comparatives, but they are often the starting point into the phenomenon of gradability in natural language semantics.

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## Limiting the scope

- Languages employ different strategies to express comparison, e.g.

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'He is taller than anyone.' (Warotamasikkkhadit 1972, via Stassen 2013)

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- However, given the amount of time we have we mostly focus on English comparatives:
  - most well-studied kinds of comparatives
  - complicated enough to keep us busy for the entirety of this introduction

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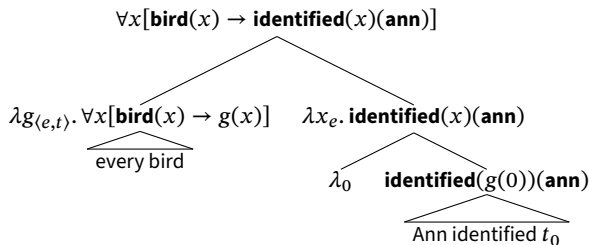
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- Specifically, I'll assume familiarity with:
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- An example of what will be presupposed:

(9) a. Ann identified every bird.

b.



## Today

- We said comparatives involve positioning two objects on a **scale**:
- (10)
- a. My dog is more **friendly** than yours.
  - b. The oboe is more **difficult** (to play) than the clarinet.
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**Side note.** Of course, predicates other than adjectives can give rise to scalar meanings:

- (11)
- a. My grandma would fish more **often** than me.
  - b. My grandma would catch more **fish** than me.
  - c. My grandma would **fish** more than me.

The degree-based approach we discuss can be extended to these cases as well.

## Part 2

# Adjectives as predicates of individuals

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- We first consider **predicative** adjectives:

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  - b. The first dinosaurs were **bipedal**.
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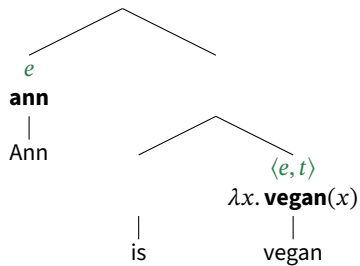
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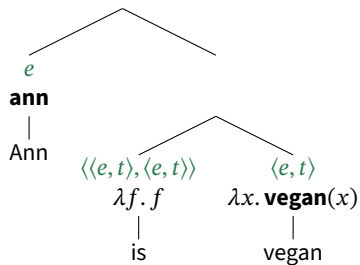
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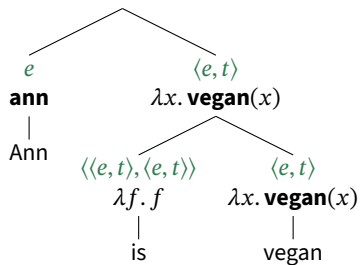
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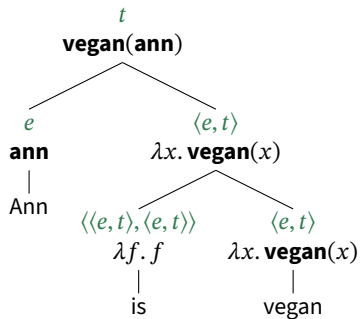
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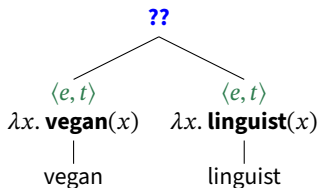
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- If adjectives are  $\langle e, t \rangle$  functions, they can't compose with nouns via functional application:

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## Interpreting attributive adjectives via Predicate Modification

- We thus introduced a new rule for interpreting complex syntactic nodes:

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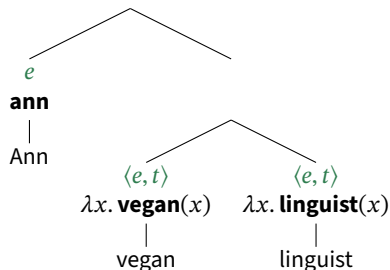
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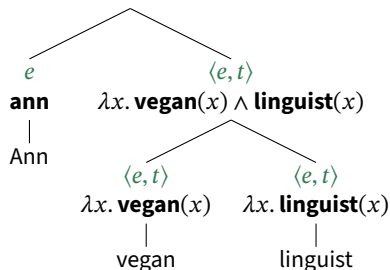
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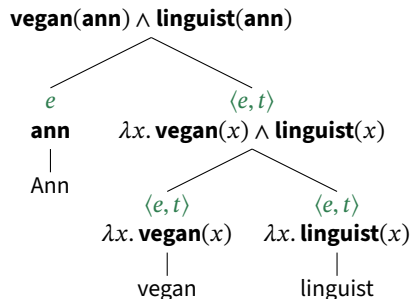
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- The analysis of adjectives as  $\langle e, t \rangle$  functions will only work for so-called **intersective** adjectives, i.e., adjectives that allow inferences such as the following:

(19) Ann is a vegan linguist.  
Ann is a birder.  
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- Many adjectives, however, are not intersective:

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- Standard analysis of these adjectives:** functions from (intensions of) NPs to NPs.

*(for a great overview of different kinds of adjectives, see Ch. 2 of Morzycki 2016)*

### Part 3

## **Problems for the basic analysis**

## Adjectives and gradability

- Predicates associated with scales are also called **gradable**.
- Gradable predicates are those that can appear in comparatives and similar constructions:

- (23)
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- Seemingly non-gradable adjectives can often be coerced into gradable ones, however:

- (25)
- a. Ann is more **vegan** than Beth.
  - b. “the earliest dinosaurs were creatures that were completely **bipedal**.”
  - c. My shoes aren’t **waterproof** enough.

## Gradable adjectives as $\langle e, t \rangle$ functions?

- Could we model gradable adjectives as  $\langle e, t \rangle$  functions?

(26) a.  $\llbracket \text{big} \rrbracket = \lambda x. \mathbf{big}(x)$   $::: \langle e, t \rangle$   
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- This seems like a non-starter:
  - an  $\langle e, t \rangle$  function is either true or false of an individual
  - we seem to need a notion of a predicate being true of something to **a certain degree**

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(from 2017 lecture notes by Roger Schwarzschild)

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- We say *tall* in its positive form is **vague**:
    - some things are definitely a heap, others definitely aren't
    - but there are things in between & boundaries are fuzzy (=borderline cases)
- ⇒ this property is emblematic of many (but not all) gradable adjectives

## Comparison classes

- Another properties of the positive form of adjectives is contextual variability:

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- A possible paraphrase the meaning of positive adjective  $\varphi$  as:
  - from a comparison class, we compute a **standard of comparison**
  - $\llbracket \varphi \rrbracket$  is true of  $x$  when  $x$  exceeds the standard of comparison in  $\varphi$ -ness

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(33) 150cm is tall for a 7yo.  
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- **Kennedy 2007:** implicit comparison

(34) Compare to Beth, Ann is tall.

- \* when Beth is 140cm and Ann is 160cm, the sentence is true
- \* when Beth is 140cm and Ann is 141cm, ...

## Part 4

# Introducing degrees & scales

## Adjectives as relations

- We now discuss a **degree-based** approach to the meaning of gradable adjectives.

(see Bartsch and Vennemann 1972, Seuren 1973, Cresswell 1976 Bierwisch 1989, Heim 1985, Stechow 1984, Kennedy 1999, among many others)

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(see Bartsch and Vennemann 1972, Seuren 1973, Cresswell 1976 Bierwisch 1989, Heim 1985, Stechow 1984, Kennedy 1999, among many others)

- In this approach, we will add **a degree argument** to gradable adjectives:

⇒ adjectives will be mapped to **relations between individuals and degrees**

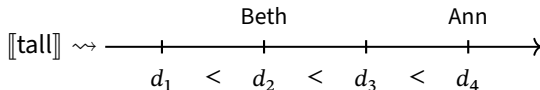
- (35)
- a.  $x \text{ is } d\text{-tall} \approx x \text{ is tall to the degree } d$
  - b.  $x \text{ is } d\text{-sharp} \approx x \text{ is sharp to the degree } d$
  - c.  $x \text{ is } d\text{-colorful} \approx x \text{ is colorful to the degree } d$

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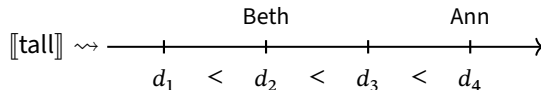
- a adjective  $\varphi$  is associated with an abstract measuring stick, i.e., a **scale**
- individuals are mapped to positions in this scale, i.e., to **degrees**
- degrees are ordered by a '**greater or equal than**' relation
- if  $x$  has more  $\varphi$ -ness than  $y$ ,  $x$  is mapped to a greater degree than  $y$  is



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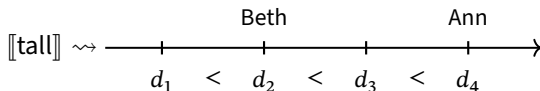
- This is all we need — and there are several ways of implementing this, e.g.

- degrees as equivalence classes of individuals (Cresswell 1976, Bale 2008, Schwarzschild 2020)
- degrees as kinds (of events) (Anderson and Morzycki 2015)
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⇒ We will make the simplest possible assumption and take degrees to be primitives.

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- |                                                                                   |                              |
|-----------------------------------------------------------------------------------|------------------------------|
| • $d \geq_S d$                                                                    | <b>reflexivity</b>           |
| • $d \geq_S d'$ or $d' \geq_S d$                                                  | <b>(strong) connectivity</b> |
| • if $d \geq_S d'$ and $d' \geq_S d$ , then $d = d'$                              | <b>antisymmetry</b>          |
| • if $d \geq_S d'$ and $d' \geq_S d''$ , then $d \geq_S d''$                      | <b>transitivity</b>          |
| • if $d >_S d'$ , then there's some $d^* \in S$ s.t. $d >_S d^*$ and $d^* >_S d'$ | <b>density</b>               |

where  $x > y := x \geq y \wedge x \neq y$

## Introducing a new type

- In a basic extensional framework, we have two basic types  $e$  and  $t$ .
- We now introduce a basic type just for degree:  $d$ .

### (36) **Semantic types**

- a.  $e$ ,  $t$  and  $d$  are semantic types.
- b. If  $\sigma$  and  $\tau$  are semantic types,  $\langle \sigma, \tau \rangle$  is a semantic type.
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- c. Nothing else is a semantic type.

- We add to our model the set of all scales  $\mathcal{S}$  and define the semantic domains as follows:

### (37) **Semantic domains**

- a.  $D_e := E$ , the set of all individuals
- b.  $D_t := \{0, 1\}$ , the set of truth values
- c.  $D_d := \{d \mid \text{for some scale } \langle S, \geq_S \rangle \in \mathcal{S}, d \in S\}$ , the set of all degrees
- d.  $D_{\langle \sigma, \tau \rangle} :=$  the set of all function from  $D_\sigma$  to  $D_\tau$